

# University Physics Quiz 6-3

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1. The Joule-Thomson coefficient is defined as  $\mu_J = (\partial T / \partial p)_H$ . It measures the temperature change during a constant enthalpy throttling process.

(a) Show by using the Legendre transformation that  $dH = TdS + Vdp$  (in a closed system) and hence  $\mu_J = V(T\alpha_p - 1)/C_p$  where  $\alpha_p = (\partial V / \partial T)_p / V$  is the thermal expansion coefficient at constant pressure.

(b) Show also that  $\mu_J = -[T(\partial p / \partial T)_V + V(\partial p / \partial V)_T] / [C_p(\partial p / \partial V)_T]$  and  $\mu_J = 0$  for an ideal gas.

(c) Show that for a mole of van der Waals fluid that  $\mu_J = \left( \frac{2a}{v} - \frac{3ab}{v^2} - bp \right) / \left[ C_p \left( p - \frac{a}{v^2} + \frac{2ab}{v^3} \right) \right]$ .

(The equation of state of a mole of van der Waals fluid is  $p = \frac{RT}{v-b} - \frac{a}{v^2}$ , where  $v$  is the molar volume,  $a$  and  $b$  are two van der Waals constants.)

Solution: (a)  $dU = TdS - pdV$ ,  $H = U + pV$ ,  $dH = dU + pdV + Vdp = TdS + Vdp$ .

$$\begin{aligned} dH &= T \left[ \left( \frac{\partial S}{\partial T} \right)_p dT + \left( \frac{\partial S}{\partial p} \right)_T dp \right] + Vdp = C_p dT + \left[ -T \left( \frac{\partial V}{\partial T} \right)_p + V \right] dp \\ &= C_p dT + V(1 - T\alpha_p) dp. \end{aligned}$$

$$(C_p = T \left( \frac{\partial S}{\partial T} \right)_p; \quad dG = -SdT + Vdp \Rightarrow \left( \frac{\partial S}{\partial p} \right)_T = - \left( \frac{\partial V}{\partial T} \right)_p.)$$

Letting  $dH = 0$ , we have  $C_p dT + V(1 - T\alpha_p) dp = 0$ , and

$$\mu_J = \left( \frac{\partial T}{\partial p} \right)_H = \frac{V(T\alpha_p - 1)}{C_p}.$$

(b) From  $0 = dH = C_p dT + \left[ -T \left( \frac{\partial V}{\partial T} \right)_p + V \right] dp$ , we have

$$\begin{aligned} \mu_J = \left( \frac{\partial T}{\partial p} \right)_H &= \frac{\left[ T \left( \frac{\partial V}{\partial T} \right)_p - V \right]}{C_p} = \frac{\left[ T \left( \frac{\partial V}{\partial T} \right)_p \left( \frac{\partial p}{\partial V} \right)_T - V \left( \frac{\partial p}{\partial V} \right)_T \right]}{C_p \left( \frac{\partial p}{\partial V} \right)_T} = \frac{\left[ -T \left( \frac{\partial p}{\partial T} \right)_V - V \left( \frac{\partial p}{\partial V} \right)_T \right]}{C_p \left( \frac{\partial p}{\partial V} \right)_T} \\ &= \frac{- \left[ T \left( \frac{\partial p}{\partial T} \right)_V + V \left( \frac{\partial p}{\partial V} \right)_T \right]}{C_p \left( \frac{\partial p}{\partial V} \right)_T}. \end{aligned}$$

For an ideal gas,  $p = \frac{nRT}{V}$ . We have  $T \left( \frac{\partial p}{\partial T} \right)_V + V \left( \frac{\partial p}{\partial V} \right)_T = T \frac{nR}{V} + V \left( -\frac{nRT}{V^2} \right) = 0$ , thus  $\mu_J = 0$ .

$$(c) \quad p = \frac{RT}{v-b} - \frac{a}{v^2}.$$

$$T \left( \frac{\partial p}{\partial T} \right)_v = \frac{RT}{v-b},$$

$$\begin{aligned}
\left(\frac{\partial p}{\partial v}\right)_T &= -\frac{RT}{(v-b)^2} + \frac{2a}{v^3} \\
\mu_J &= \frac{-\left[T\left(\frac{\partial p}{\partial T}\right)_v + v\left(\frac{\partial p}{\partial v}\right)_T\right]}{C_p\left(\frac{\partial p}{\partial v}\right)_T} = -\frac{\left[\frac{RT}{v-b} - \frac{RTv}{(v-b)^2} + \frac{2a}{v^2}\right]}{C_p\left[-\frac{RT}{(v-b)^2} + \frac{2a}{v^3}\right]} = -\frac{\left[\frac{RT}{v-b} - \frac{RTv}{(v-b)^2} + \frac{2a}{v^2}\right](v-b)}{C_p\left[-\frac{RT}{(v-b)^2} + \frac{2a}{v^3}\right](v-b)} \\
&= -\frac{\left[RT - \frac{RTv}{v-b} + \frac{2a}{v^2}(v-b)\right]}{C_p\left[-\frac{RT}{(v-b)} + \frac{2a}{v^2} - \frac{2ab}{v^3}\right]} = -\frac{\left[\frac{RT(v-b)}{v-b} - \frac{RTv}{v-b} + \frac{2a}{v^2}(v-b)\right]}{C_p\left(\frac{a}{v^2} - \frac{2ab}{v^3} - p\right)} = -\frac{\left[-b\frac{RT}{v-b} + \frac{2a}{v^2}(v-b)\right]}{C_p\left(\frac{a}{v^2} - \frac{2ab}{v^3} - p\right)} \\
&= -\frac{\left[-b\left(\frac{RT}{v-b} - \frac{a}{v^2}\right) + \frac{2a}{v} - \frac{3ab}{v^2}\right]}{C_p\left(\frac{a}{v^2} - \frac{2ab}{v^3} - p\right)} = \frac{\left(bp + \frac{3ab}{v^2} - \frac{2a}{v}\right)}{C_p\left(\frac{a}{v^2} - \frac{2ab}{v^3} - p\right)} = \frac{\left(\frac{2a}{v} - \frac{3ab}{v^2} - bp\right)}{C_p\left(p - \frac{a}{v^2} + \frac{2ab}{v^3}\right)}.
\end{aligned}$$

2. By using the fundamental equation of thermodynamics, show that

$$(a) \quad \left(\frac{\partial C_p}{\partial p}\right)_{T,n} = -T\left(\frac{\partial^2 V}{\partial T^2}\right)_{p,n}, \quad (b) \quad V\left(\frac{\partial p}{\partial \mu}\right)_{T,n} = n.$$

(b) In an ideal gas of molecular mass  $m$ , it is found that the speeds  $v_1$  and  $v_2$  have the equal value of the Maxwell speed distribution function. Find the temperature of the gas.

Solution: (a) The fundamental equation of thermodynamics reads

$$TdS = dU + pdV - \mu dn = dH - Vdp - \mu dn \quad \text{or} \quad dH = TdS + Vdp + \mu dn.$$

$$\text{Thus} \quad C_p = \left(\frac{\partial H}{\partial T}\right)_{p,n} = T\left(\frac{\partial S}{\partial T}\right)_{p,n}.$$

$$\text{From } dG = d(H - TS) = -SdT + Vdp + \mu dn, \text{ we produce a Maxwell relation } \left(\frac{\partial S}{\partial p}\right)_{T,n} = -\left(\frac{\partial V}{\partial T}\right)_{p,n}.$$

$$\text{Therefore, } \left(\frac{\partial C_p}{\partial p}\right)_{T,n} = -T\left[\frac{\partial}{\partial p}\left(\frac{\partial S}{\partial T}\right)_{p,n}\right]_{T,n} = -T\left[\frac{\partial}{\partial T}\left(\frac{\partial S}{\partial p}\right)_{T,n}\right]_{p,n} = -T\left(\frac{\partial^2 V}{\partial T^2}\right)_{p,n}.$$

$$(b) \quad \text{From } dG = -SdT + Vdp + \mu dn, \text{ we have } V = \left(\frac{\partial G}{\partial p}\right)_{T,n}.$$

$$\text{Since } G(T, p, \lambda n) = \lambda G(T, p, n), \text{ we have the Euler's relation } G = \left(\frac{\partial G}{\partial n}\right)_{T,p} n = \mu n.$$

$$\text{Therefore, } V = \left(\frac{\partial(\mu n)}{\partial p}\right)_{T,n} = n\left(\frac{\partial \mu}{\partial p}\right)_{T,n} \quad \text{or} \quad V\left(\frac{\partial p}{\partial \mu}\right)_{T,n} = n.$$

3. A definite amount of gas at temperature  $T_0$  initially is confined to volume  $V_0$  in a partitioned insulated rigid cylinder. The unoccupied part, also with volume  $V_0$  is initially evacuated. Suppose the gas expands spontaneously and freely into the unoccupied part, and eventually comes to the equilibrium.

(a) What is the change in internal energy in the process?

(b) Show that the final temperature of the gas is

$$T_f = T_0 + \int_{V_0}^{2V_0} \frac{1}{C_V} \left[ p - T \left( \frac{\partial p}{\partial T} \right)_V \right] dV.$$

(c) Show that  $T_f = T_0$  for an ideal gas and  $T_f < T_0$  for a gas that obeys the van der Waals equation. Explain the results in terms of the kinetic theory of molecules.

Solution: (a) In the process, no work is done and no heat is transferred. Thus the internal energy is unchanged,  $\Delta U = 0$ .

To deal with the problem, we consider the quantity  $(\partial T / \partial V)_U$ . Using the cyclic chain rule, we have

$$\left( \frac{\partial T}{\partial V} \right)_U = - \frac{\left( \frac{\partial U}{\partial V} \right)_T}{\left( \frac{\partial U}{\partial T} \right)_V} = - \frac{\left( \frac{\partial U}{\partial V} \right)_T}{C_V}.$$

From

$$dU = TdS - pdV = T \left( \frac{\partial S}{\partial T} \right)_V + \left[ T \left( \frac{\partial S}{\partial V} \right)_T - p \right] dV,$$

we have

$$\left( \frac{\partial U}{\partial V} \right)_T = T \left( \frac{\partial S}{\partial V} \right)_T - p.$$

From  $dF = -SdT - pdV$ , we have the Maxwell relation  $\left( \frac{\partial S}{\partial V} \right)_T = \left( \frac{\partial p}{\partial T} \right)_V$ . Then, we have

$$\left( \frac{\partial U}{\partial V} \right)_T = T \left( \frac{\partial p}{\partial T} \right)_V - p.$$

Thus, we have

$$dT = \left( \frac{\partial T}{\partial V} \right)_U dV + \left( \frac{\partial T}{\partial U} \right)_V dU = \left( \frac{\partial T}{\partial V} \right)_U dV = \frac{1}{C_V} \left( p - T \left( \frac{\partial p}{\partial T} \right)_V \right) dV,$$

and

$$\int_{T_0}^{T_f} dT = \int_{V_0}^{2V_0} \frac{1}{C_V} \left[ p - T \left( \frac{\partial p}{\partial T} \right)_V \right] dV, \quad T_f = T_0 + \int_{V_0}^{2V_0} \frac{1}{C_V} \left[ p - T \left( \frac{\partial p}{\partial T} \right)_V \right] dV$$

(c) For an ideal gas

$$p - T \left( \frac{\partial p}{\partial T} \right)_V = \frac{nRT}{V} - T \frac{nR}{V} = 0,$$

Then  $T_f = T_0$ .

For the van der Waals gas,  $p = \frac{nRT}{V-nb} - \frac{n^2 a}{V^2}$ ,

$$p - T \left( \frac{\partial p}{\partial T} \right)_V = \frac{nRT}{V-nb} - \frac{n^2 a}{V^2} - T \frac{nR}{V-nb} = -\frac{n^2 a}{V^2} < 0,$$

Then  $T_f < T_0$ .

For the ideal gas, as the gas expands, the kinetic energy of the molecules is unchanged and so is the temperature.

For the van der Waals gas, as the gas expands, the spacing among molecules increases. Thus the attractive (negative) interaction potential energy increases and the translational kinetic energy of the molecules decreases and therefore the temperature drops.

4. (a) There is an amount of matter in gas-liquid equilibrium. Assume that the gas phase of the matter behaves as an ideal gas and the volume of the matter in liquid phase is negligible. Show that the expansion coefficient of the matter in the gas-liquid equilibrium is given by

$$\frac{1}{V} \frac{dV}{dT} = \frac{1}{T} \left( 1 - \frac{L}{RT} \right)$$

where  $L$  is the molar latent heat.

(b) Show that the change in molar internal energy in a phase transition can be expressed as

$$\Delta u = L \left( 1 - \frac{p}{T} \frac{dT}{dp} \right).$$

Solution: (a) The volume of the matter is approximately the volume of the part of matter in gas phase. Then, the molar volume of the matter can be written as  $v = v(T, p)$ . Thus

$$\frac{1}{v} \frac{dv}{dT} = \frac{1}{v} \left( \left( \frac{\partial v}{\partial T} \right)_p + \left( \frac{\partial v}{\partial p} \right)_T \frac{dp}{dT} \right). \quad (1)$$

Since the gas phase of the matter behaves as an ideal gas, then  $pv = xRT$ , where  $x$  is the molar fraction of the matter in gas phase. Therefore

$$\frac{1}{v} \left( \frac{\partial v}{\partial T} \right)_p = \frac{1}{T}, \quad \frac{1}{v} \left( \frac{\partial v}{\partial p} \right)_T = -\frac{1}{p}. \quad (2)$$

Using the Clapeyron equation,  $\frac{dp}{dT} = \frac{L}{T(V_m^g - V_m^l)} \approx \frac{L}{TV_m^g}$ . Noting that  $V_m^g = \frac{RT}{p}$ , we have

$$\frac{dp}{dT} = \frac{Lp}{RT^2}. \quad (3)$$

Substituting (2) and (3) into (1), we finally obtain

$$\frac{1}{v} \frac{dv}{dT} = \frac{1}{T} \left( 1 - \frac{L}{RT} \right).$$

(b) The condition for the phase equilibrium is

$$\mu_\alpha(T, p) = \mu_\beta(T, p)$$

where  $\mu$  represents the chemical potentials of the two phases.

Since  $\mu = u + pv - Ts$  where  $v$  is the molar volume and  $s$  is the molar entropy, we have

$$u_\alpha + pv_\alpha - Ts_\alpha = u_\beta + pv_\beta - Ts_\beta.$$

Thus  $\Delta u = u_\beta - u_\alpha = T(s_\beta - s_\alpha) - p(v_\beta - v_\alpha) = L - p(v_\beta - v_\alpha)$ .

Using the Clausius-Clapeyron equation  $\frac{dp}{dT} = \frac{L}{T(v_\beta - v_\alpha)}$ , we have  $v_\beta - v_\alpha = \frac{L}{T} \frac{dT}{dp}$ .

Finally, we obtain  $\Delta u = L \left( 1 - \frac{p}{T} \frac{dT}{dp} \right)$ .